

Organic Chemistry Practical 1: Qualitative Analysis – Functional Group Identification

Systematic Diagnostic Test-Tube Reactions of Alcohols, Carbonyls, Carboxylic Acids & Halogenoalkanes

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Organic Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm³; balance readings to 0.01 g / 0.001 g.
 - Thermometer readings to the nearest 0.5 °C.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Diagnostic Differentiation of Alcohols, Aldehydes, and Ketones: FA 1, FA 2, and FA 3

Cambridge Syllabus Reference: 9701/33/M/J/22/Q3 · [20 Marks]

Qualitative Analysis Instructions:

You are provided with three unlabelled organic liquids: FA 1, FA 2, and FA 3.

Each liquid is one of the following compounds: propan-1-ol (primary alcohol), propanal (aldehyde), or propanone (ketone).

Reagents Provided:

- Acidified potassium dichromate(VI), $K_2Cr_2O_7$ / dilute H_2SO_4
- Tollens' reagent (prepared freshly from $AgNO_3$, dilute $NaOH$, and dilute NH_3)
- Fehling's solution (equal volumes of Fehling's A and Fehling's B)
- 2,4-dinitrophenylhydrazine (2,4-DNPH / Brady's reagent)
- Dry sodium metal pieces (stored under paraffin oil).

Safety Precautions:

- Organic compounds and 2,4-DNPH are flammable and harmful. 2,4-DNPH stains skin and clothing.
- NEVER heat organic mixtures directly over an open Bunsen burner flame. Always use a hot water bath.

(a) Carry out the specified diagnostic tests on FA 1, FA 2, and FA 3. Record all observations in the table below. [8]

Test	Observations	Deductions
Brady's Reagent Test: To 5 drops of organic sample, add 1 cm ³ of 2,4-DNPH reagent.		
Tollens' Silver Mirror Test: To 1 cm ³ of fresh Tollens' reagent, add 5 drops of organic sample and warm in a water bath at 60°C for 5 minutes.		
Fehling's Alkaline Copper Test: To 1 cm ³ of mixed Fehling's solution, add 5 drops of organic sample and warm in a boiling water bath for 3 minutes.		
Dichromate Oxidation Test: To 1 cm ³ of acidified $K_2Cr_2O_7$, add 5 drops of organic sample and warm gently in a water bath.		
Sodium Metal Test: To 1 cm ³ of dry organic liquid in a dry test-tube, add a small freshly cut piece of sodium metal.		

(b) From your experimental observations, identify the chemical identity of FA 1, FA 2, and FA 3 and provide concise justifications for each deduction. [6]

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(c) Write the balanced chemical equation for the reaction of propan-1-ol (FA 1) with sodium metal. Name the organic product formed. [2]

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(d) State the identity of the transition metal species responsible for the color change from orange to green when propanal is warmed with acidified potassium dichromate(VI), giving its oxidation number. [2]

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(e) Describe how the silver mirror formed in the Tollens' test on propanal should be safely removed and cleaned from the glassware at the end of the practical session. [2]

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Question 2: Identification of Halogenoalkanes and Carboxylic Acids: FA 4, FA 5, and FA 6

Cambridge Syllabus Reference: 9701/34/O/N/23/Q3 · [20 Marks]

Qualitative Analysis Instructions:

You are provided with three liquids: FA 4, FA 5, and FA 6.

Each liquid is one of the following: ethanoic acid, 1-chlorobutane, or 1-bromobutane.

Reagents Provided:

- Solid sodium hydrogencarbonate, $\text{NaHCO}_3(\text{s})$
- Aqueous sodium hydroxide, $\text{NaOH}(\text{aq})$, 2.0 mol dm^{-3}
- Dilute nitric acid, HNO_3 , 2.0 mol dm^{-3}
- Aqueous silver nitrate, AgNO_3 , 0.05 mol dm^{-3}
- Dilute aqueous ammonia, NH_3 , 1.0 mol dm^{-3}
- Concentrated aqueous ammonia, conc. NH_3 , 8.0 mol dm^{-3} (fume cupboard)
- Fresh limewater, $\text{Ca}(\text{OH})_2(\text{aq})$.

(a) Perform the diagnostic tests specified below and complete the observation table. [8]

Test	Observations	Deductions
Bicarbonate Test: To 1 cm depth of FA 4, FA 5, and FA 6 in separate tubes, add a spatula-tip of solid NaHCO_3 . Test any evolved gas with limewater.		
Alkaline Hydrolysis & Halide Test: To 1 cm depth of FA 5 and FA 6 in separate tubes: (i) Add 2 cm depth of $\text{NaOH}(\text{aq})$ and heat in a boiling water bath for 5 minutes. (ii) Cool, acidify with dilute HNO_3 until acidic to litmus. (iii) Add 5 drops of aqueous AgNO_3 .		
Ammonia Confirmation Test: To the precipitates obtained from the hydrolysis of FA 5 and FA 6: (i) Add excess dilute aqueous NH_3 . (ii) If insoluble, add concentrated aqueous NH_3 .		

(b) Deduce the identity of FA 4, FA 5, and FA 6. [3]

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(c) Write an ionic equation, with state symbols, for the reaction between ethanoic acid and hydrogencarbonate ions in the bicarbonate test. [2]

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(d) Explain why the hydrolysis mixture must be acidified with dilute nitric acid BEFORE adding aqueous silver nitrate. [2]

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(e) Compare the relative rates of hydrolysis of 1-chlorobutane (FA 5) and 1-bromobutane (FA 6). Explain this difference in rate by reference to bond parameters from the Cambridge Data Booklet. [4]

(f) Write the general ionic equation for the nucleophilic substitution hydrolysis of 1-bromobutane with hydroxide ions. [1]

Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Organic Chemistry Practical 1: Qualitative Analysis – Functional Group Identification

Question 1: Diagnostic Differentiation of Alcohols, Aldehydes, and Ketones: FA 1, FA 2, and FA 3 [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	2,4-DNPH test: FA 1 gives no precipitate / remains yellow solution; FA 2 and FA 3 give bright orange-yellow precipitate [2]. Tollens' test: FA 1 and FA 3 show no reaction / solution remains colorless; FA 2 produces a shiny silver mirror on the tube walls (or dark grey ppt) [2]. Fehling's test: FA 2 turns from deep royal blue to form a brick-red / orange-brown precipitate of Cu ₂ O; FA 1 and FA 3 remain deep blue [1]. Acidified dichromate test: FA 1 and FA 2 turn from orange to green (Cr ³⁺); FA 3 shows no reaction / remains orange [2]. Sodium metal test: FA 1 effervesces / produces bubbles of H ₂ gas; FA 2 and FA 3 show no gas evolution [1].	[8]
(b)	FA 1 is propan-1-ol [1]: negative 2,4-DNPH (not a carbonyl), oxidized by acidified dichromate (turns green), effervesces with sodium metal (contains -OH group) [1]. FA 2 is propanal [1]: positive 2,4-DNPH (contains C=O carbonyl), oxidized by Tollens' and Fehling's (aldehyde group -CHO) [1]. FA 3 is propanone [1]: positive 2,4-DNPH (carbonyl), but negative Tollens', Fehling's, and dichromate tests (ketone resist oxidation under mild conditions) [1].	[6]
(c)	Equation: $2\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + 2\text{Na} \rightarrow 2\text{CH}_3\text{CH}_2\text{CH}_2\text{ONa} + \text{H}_2$ (or $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{Na} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{ONa} + 1/2 \text{H}_2$) [1]. Name: Sodium propoxide [1].	[2]
(d)	Species: Chromium(III) ion / Cr ³⁺ (aq) [1]; Oxidation number: +3 (reduced from +6 in Cr ₂ O ₇ ²⁻) [1].	[2]
(e)	Dissolve the silver mirror with dilute nitric acid (HNO ₃) in a fume cupboard [1]; Silver fulminate (explosive) can form if Tollens' residues dry out or are left standing, so all solutions must be rinsed immediately with copious water down the sink [1].	[2]

Question 2: Identification of Halogenoalkanes and Carboxylic Acids: FA 4, FA 5, and FA 6 [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	NaHCO ₃ test: FA 4 gives vigorous effervescence / bubbling, gas turns limewater milky; FA 5 and FA 6 show no reaction / no bubbling [2]. Hydrolysis + AgNO ₃ test: FA 5 gives a white precipitate [1]; FA 6 gives a cream precipitate [1]. Relative rate: Precipitate forms noticeably faster in FA 6 than in FA 5 [1]. Ammonia test: FA 5 white precipitate dissolves completely in dilute aqueous ammonia [1]; FA 6 cream precipitate is insoluble in dilute ammonia but dissolves in concentrated ammonia [2].	[8]
(b)	FA 4 is ethanoic acid [1]; FA 5 is 1-chlorobutane [1]; FA 6 is 1-bromobutane [1].	[3]
(c)	$\text{CH}_3\text{COOH}(\text{l or aq}) + \text{HCO}_3^-(\text{aq or s}) \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ [2] (1 mark for correct formulae, 1 mark for state symbols).	[2]
(d)	To neutralize the unreacted excess sodium hydroxide (OH ⁻ ions) [1]; otherwise OH ⁻ ions would react with Ag ⁺ to precipitate brown silver(I) oxide (Ag ₂ O), masking or giving a false test for halide ions [1].	[2]
(e)	1-bromobutane hydrolyzes faster than 1-chlorobutane (cream precipitate appears more rapidly) [1]. The rate of nucleophilic substitution is determined by the strength / bond enthalpy of the carbon-halogen bond (C-X), not bond polarity [1]. The C-Br bond enthalpy (276 kJ mol ⁻¹) is lower than the C-Cl bond enthalpy (338 kJ mol ⁻¹) [1]. Because bromine has a larger atomic radius than chlorine, orbital overlap with carbon is poorer, so less activation energy is required to break the C-Br bond [1].	[4]
(f)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} + \text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + \text{Br}^-$ [1].	[1]